

# LESSON-4

## MOLE CONCEPT

$$\text{NO. OF MILLIMOLES} = M \cdot V$$

$$m = \frac{M \times 1000}{1000 d - M \cdot Wt \cdot M \text{ Solute}}$$

( $\therefore$  Molarity  $\nabla$ s Molality)

$$M \times n \text{ factor} = N \text{ Normality}$$

$$\text{Parts Per million PPM} = \frac{W}{W} \cdot 10^4$$

$$\frac{M_1 V_1}{1} = \frac{M_2 V_2}{2}$$

$$\frac{N_1 V_1}{1} = \frac{N_2 V_2}{2}$$

$$i. M = \frac{M_1 V_1 + M_2 V_2}{V_{\text{result}}}$$

$$ii. N = \frac{N_1 V_1 + N_2 V_2}{V_{\text{result}}}$$

Molarity/Molality  $\nabla$ s Mole fraction:

$$i. m = \frac{x_{\text{Solute}} \times 1000}{(1 - x_{\text{Solute}}) \times M_{\text{Solvent}}}$$

$$ii. M = \frac{1000 d \times x_{\text{Solute}}}{M \times x_{\text{Solute}} + M \times x_{\text{Solvent}}}$$

$$M = \frac{1000 d \times x_{\text{Solute}}}{M \times x_{\text{Solute}} + M \times x_{\text{Solvent}}}$$

$$= \frac{\text{Mol. Mass (g)} \cdot n \text{ factor}}{\text{Eq. wt in g}}$$

$$i. \text{No. of Gram Equivalents} = \frac{Wt}{\text{Eq. wt.}}$$

$$ii. \text{Gram Equivalent}$$

$$N = \frac{Wt_{\text{Solute}}}{M_{\text{wt Solute}}} \times \frac{n \times 1000}{V_{\text{ml Solvent}}}$$

$$\text{Density} = \frac{\text{Mass}}{V_{\text{ol}}}$$

$$N = \text{No. of Gram Equivalents}$$

$$\text{Mole} = \frac{Wt}{M_{\text{wt}}}$$

$V_{\text{lt.}}$

$$i. \text{Molarity (M)} = \frac{Wt_{\text{Solute}}}{M_{\text{wt Solute}}} \times \frac{1000}{V_{\text{ml Solution}}}$$

$$ii. \text{Molality (m)} = \frac{Wt_{\text{Solute}}}{M_{\text{wt Solute}}} \times \frac{1000}{Wt_{\text{Solvent}}}$$

i. Molecular Mass of Gas =  $2 \cdot (V.D)$

$$V.D = \frac{\text{Molecular Mass of Gas}}{2}$$

ii. Average Molecular Mass :  $\frac{n_A M_A + n_B M_B + n_C M_C + \dots}{n_A + n_B + n_C + \dots}$

where.

$n_A, n_B, n_C, \dots$  are Moles of Gases

$M_A, M_B, M_C, \dots$  are Molecular Masses of Gas.

~~PV~~  $PV = nRT$

(ml)   
 (Atm) or Kelvin   
  $\uparrow \uparrow \uparrow$

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2} \quad (\because \text{Here } P_1, V_1, T_1 \text{ are STP Conditions})$$

$$\text{No. of Moles} = \frac{\text{Vol}}{\frac{\text{Molar Vol of Gas} \times \text{Pressure}}{(\text{At given Temp})}}$$

$$1 \text{ amu} = 1.66 \times 10^{-24} \text{ g} = 1 \text{ Dalton} = 1 \text{ Avogaram} = 1.66 \times 10^{-27} \text{ kg}$$

$$\text{Avagadro's No. } 6.023 \times 10^{23}$$

$$\text{RAM} = \frac{\text{Mass of Atom}}{\text{Standard Atom}} \quad (\because \text{NO Units})$$

$$\text{RMM} = \frac{\text{Molecular Mass of one Molecule}}{1 \text{ amu.}}$$

$$\text{Molecular Mass} = \frac{\text{Mass of one Molecule}}{1 \text{ amu.}}$$

$n =$  Molecular Mass formula   
 Empirical Mass formula.

$$\% \text{ Purity} = \frac{\text{Amount of Pure}}{\text{Total Amount of Impure}} \times 100 \quad \boxed{\% \text{ Impurity} = 100 - \% \text{ Purity}}$$

## Concentrations:

1. Weight Percentage (w%):  $\frac{\text{wt of Solute}}{\text{wt of Solution}} \times 100$
2. Volume Percentage (v%):  $\frac{\text{Vol of Solute (ml)}}{\text{Vol of Solution (ml)}} \times 100$
3. Weight Vol:  $\frac{\text{wt of Solute}}{\text{Vol of Solution}} \times 100$
4. Molarity =  $\frac{\text{w\% density} \cdot 10}{(\text{Mwt})_{\text{solute}}}$  Molar, Mole/Litre.
5. Molality Mol/kg (or) Molal.
6. Normality (N): Normal/Equivalents Per litre.

Normal IN      Normality = Molarity  $\times$  Basicity of acid.  
Seminormal  $\frac{N}{2}$   
Decinormal  $\frac{N}{10}$   
Centinormal  $\frac{N}{100}$   
Milli Normal  $\frac{N}{1000}$

## 7. Mole fraction:

$$X_A = \frac{n_A}{n_A + n_B} \quad X_B = \frac{n_B}{n_A + n_B} \quad \boxed{X_A + X_B = 1}$$

## Hardness of water:

$$\frac{\text{wt of CaCO}_3}{\text{wt of hard water}} \times 10 \text{ ppm.}$$

$$\% \text{ Yield} = \frac{\text{Actual Yield}}{\text{Theoretical Yield}} \times 100$$

$$\% \text{ Crystallisation of water} = \frac{\text{Weight of water in 1 Mole of Salt}}{\text{Molar Mass of Salt}} \times 100$$

$$\text{Equivalent Mass} = \frac{\text{At. Mass}}{n\text{-factor}} = \frac{\text{At. wt.}}{\text{Valency}}$$



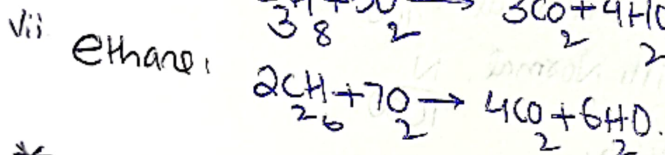
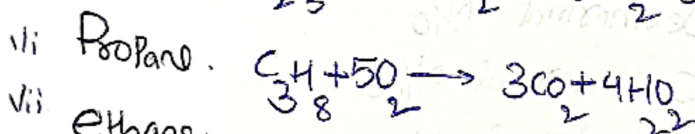
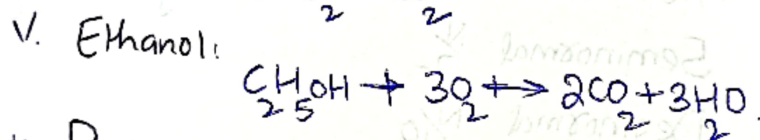
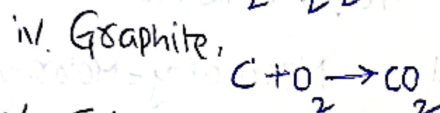
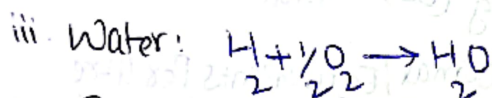
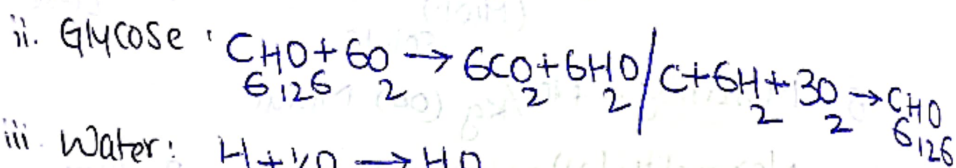
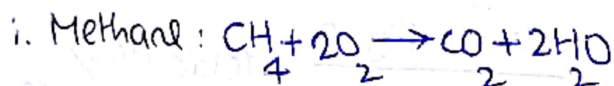
: Percentage Element:

$$\% = \frac{\text{No of Atoms} \times \text{Atomic wt}}{\text{Total compound wt}} \times 100.$$

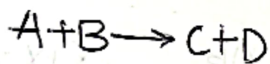
: Relation B/w molality/Molarity (2024 Direct P4Q's)

$$m = \frac{1000 M}{1000d - (M \times M_w \text{ solute})}$$

Combustion (Addition of  $O_2$ )



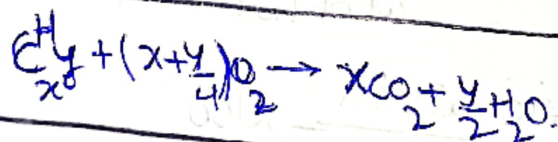
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where:

A B C D of Different Weight- but

$$x \text{ Moles of A} = y \text{ Moles of C}$$



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